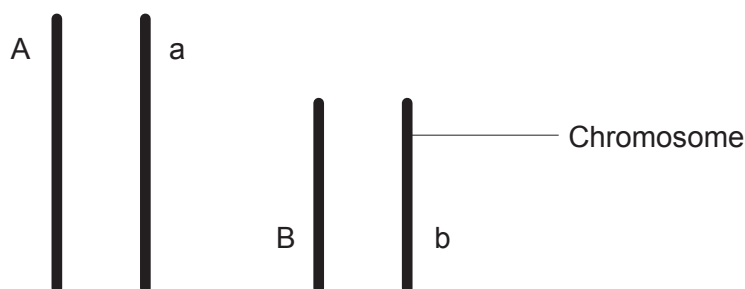


5. (a) An organism heterozygous for two genes, A and B, has the genotype AaBb.

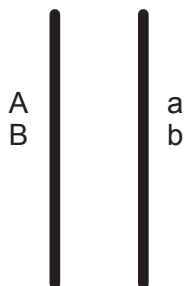
- (i) The alleles are arranged as shown in the diagram below. State the genotypes of the gametes produced and the expected phenotype ratio of the offspring in the F_1 following a cross between AaBb and AaBb. [2]



Gamete genotypes:

F_1 phenotype ratio:

- (ii) The alleles could be arranged as shown below.



Explain why a cross between AaBb and AaBb in the above could produce a phenotype ratio of 3:1. [2]

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- (b) Explain why two individuals, both with genotype AaBb, may not be phenotypically identical for these characteristics. [1]

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